

Kirana Connect - Technology Stack

1. Backend API (Node.js & Express)

- express

Web framework to create API routes and handle HTTP requests. Usage: `const app = express(); app.get(...);`

- pg (node-postgres)

PostgreSQL client to connect to the DB and securely manage data. Usage: `const pool = new Pool();`

- socket.io

Real-time library for instant order updates between apps. Usage: `io.on("connection", ...)`

- jsonwebtoken (JWT)

For secure authentication across API requests. Usage: `jwt.sign(payload, secret)`

- bcryptjs

To hash passwords securely in the database. Usage: `bcrypt.hash(password, 10)`

- multer & cloudinary

Multer handles file uploads, Cloudinary handles cloud storage for product images.

- zod

Schema validation for ensuring incoming API data is correct.

2. Frontend Applications (Customer & Store Apps)

- expo

Framework over React Native to build for Android, iOS, and Web. Usage: `npx expo start`

- expo-router

File-based routing for managing screen navigation. Usage: Create files in `app/` directory.

- react-native-maps & expo-location

Mapping libraries for GPS coordinates and visual store map.

- axios

HTTP client to fetch data from our Node.js backend. Usage: `axios.get(...)`

- zustand

State-management library to share data across screens. Usage: `create((set) => (...))`

- @expo/vector-icons

Icon library for the user interfaces. Usage: `<Ionicons name="cart" />`